Marwadi University Marwadi Chandarana Group NAAC A+	Marwadi University Faculty of Department of Information and Communication Technology	
Digital Signal and Image Processing (01CT1513)	Aim: Perform Gray Level Operation Image	
Experiment : 7	Date: 07/08/2026	Enrolment No: 92400133125

Aim : Perfrom Gray Level operation Image

GitHub :
https://github.com/mankumarbhalodiya/Digital-Signal-and-Image-Processing/tree/main/Experiment%207

Code :

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read image in grayscale
src_image = cv2.imread("1.jpg", cv2.IMREAD_GRAYSCALE)

# Check if image is loaded
if src_image is None:
    print("Image not found!")
    exit()

# -------------------------
# 1. Image Negation
# -------------------------
negative_image = 255 - src_image

# -------------------------
# 2. Thresholding
# -------------------------
_, thresholded_image = cv2.threshold(src_image, 128, 255,
cv2.THRESH_BINARY)

# -------------------------
# 3. Gamma Correction
# -------------------------
gamma = 1.5

# Normalize image to range [0,1]
normalized_image = src_image / 255.0

# Apply gamma correction
```

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```
gamma_corrected_image = np.power(normalized_image, 1 / gamma)

# Convert back to range [0,255]
gamma_corrected_image = np.uint8(gamma_corrected_image * 255)

# -------------------------
# Display Images
# -------------------------
plt.figure(figsize=(15, 5))

plt.subplot(1, 4, 1)
plt.imshow(src_image, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1, 4, 2)
plt.imshow(negative_image, cmap="gray")
plt.title("Negative")
plt.axis("off")

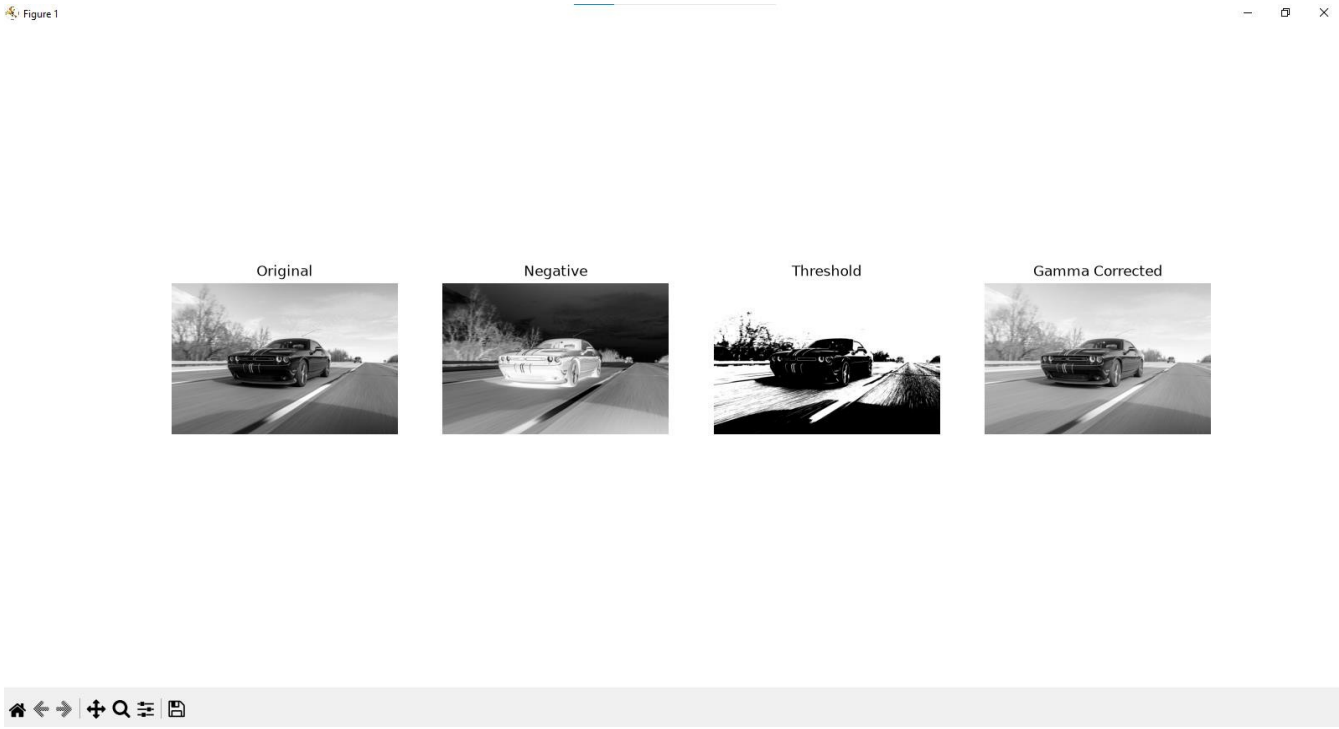
plt.subplot(1, 4, 3)
plt.imshow(thresholded_image, cmap="gray")
plt.title("Threshold")
plt.axis("off")

plt.subplot(1, 4, 4)
plt.imshow(gamma_corrected_image, cmap="gray")
plt.title("Gamma Corrected")
plt.axis("off")

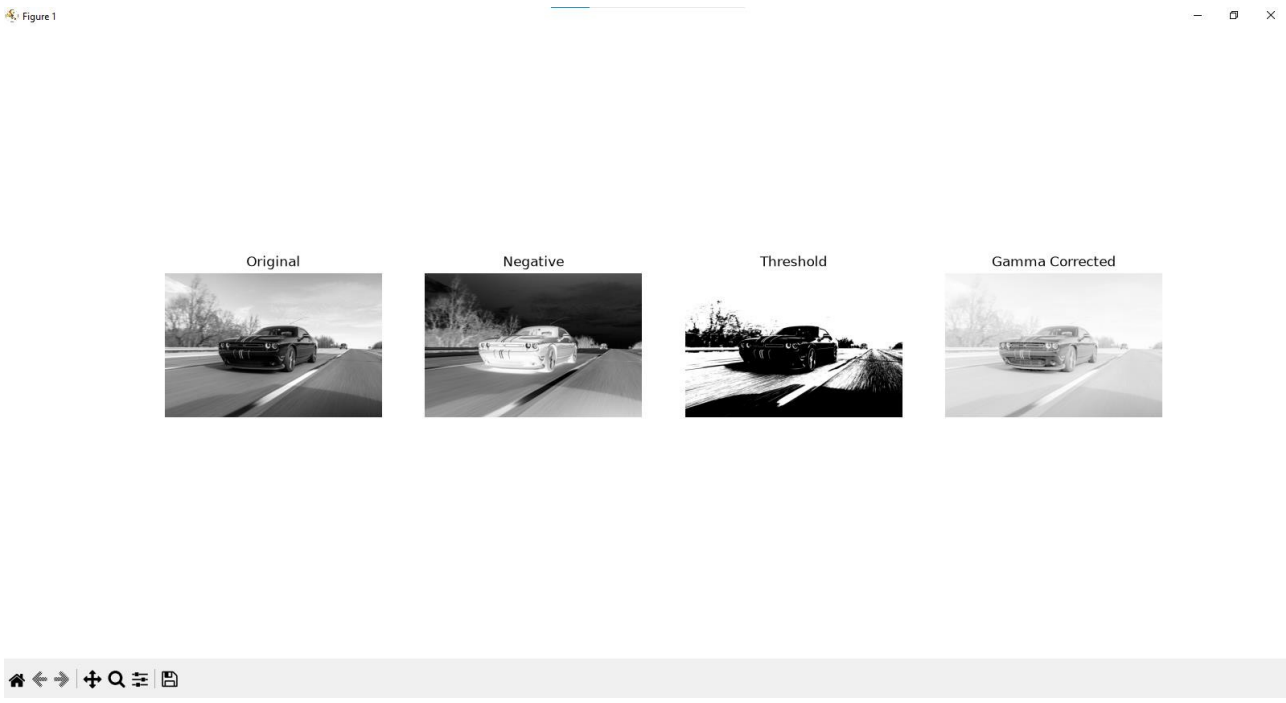
plt.show()
```

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Output :



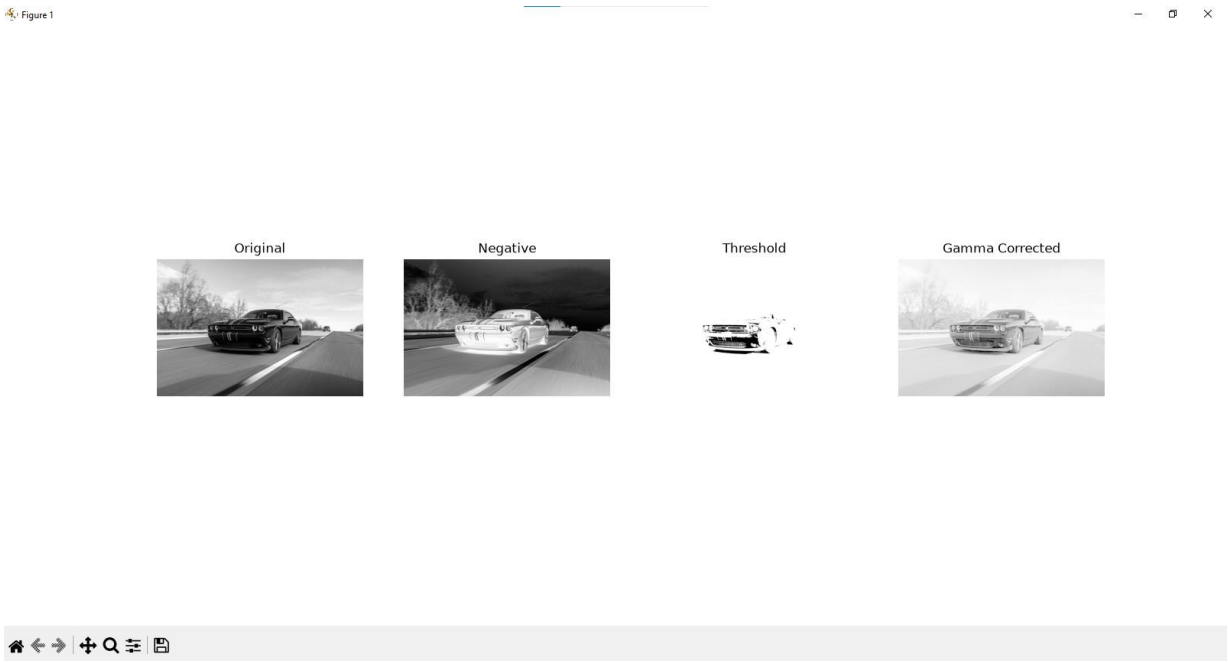
Gama Value = 5



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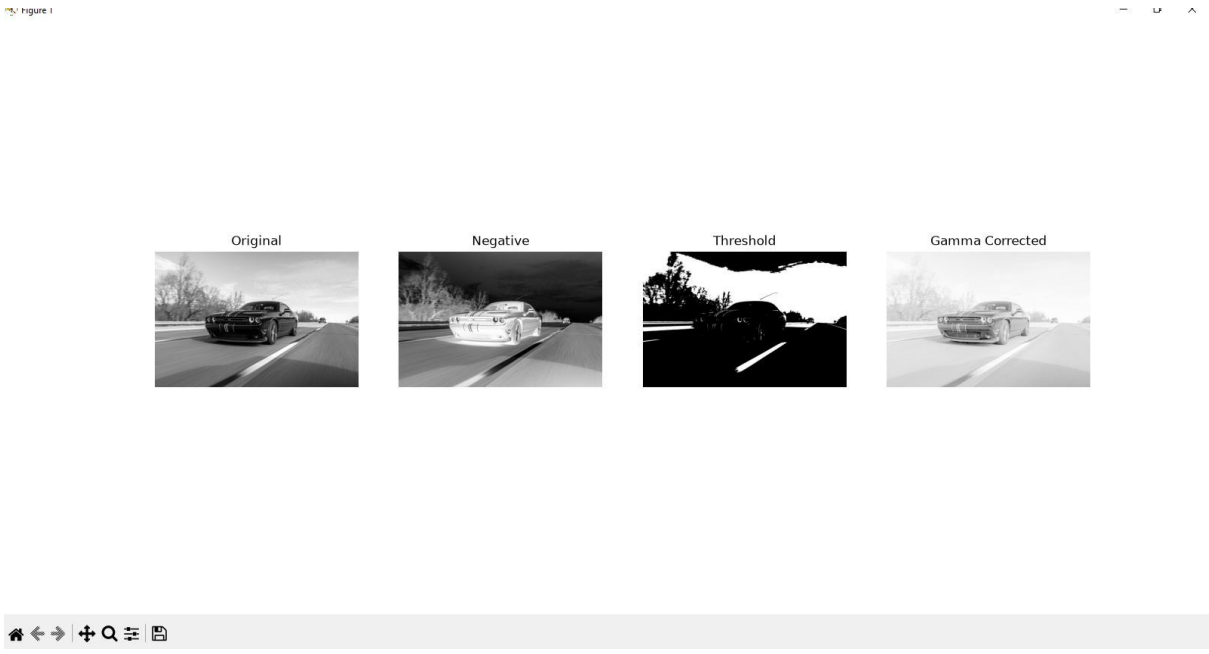
Thresold Value = 10

```
_, thresholded_image = cv2.threshold(src_image, 10, 255, cv2.THRESH_BINARY)
```



Thresold Value : 200

```
_, thresholded_image = cv2.threshold(src_image, 200, 255, cv2.THRESH_BINARY)
```



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Conclusion :

The program successfully performs basic image processing operations on a grayscale image. It creates a negative image, converts the image into black and white using thresholding, and adjusts the brightness using gamma correction. These techniques help improve image appearance and make important features easier to observe. This experiment demonstrates how simple image processing methods can be applied using Python, OpenCV, and NumPy.